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(54) **EXERCISE SAFETY PREDICTION BASED ON PHYSIOLOGICAL CONDITIONS**

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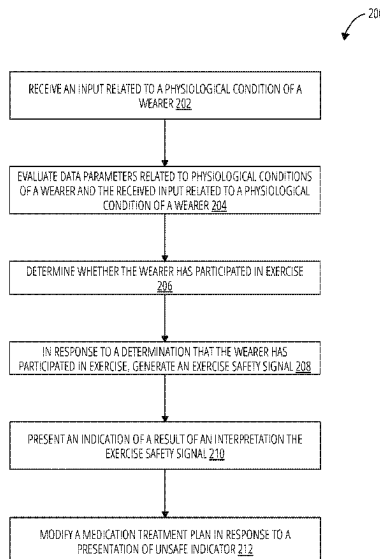
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(57) **ABSTRACT**

Described are techniques, processes, devices, computer-readable media that enable provision of an indication of whether it is safe for a person with diabetes to participate in exercise while using a wearable drug delivery system. A processor may receive or obtain physiological data related to a condition of a wearer of the wearable drug delivery system and by evaluating an exercise model that uses inputs related to the physiological data to make the determination of whether it is safe to exercise and output an exercise safety signal. Modifications to the wearer's medication treatment plan and other actions may be based on an outputted exercise safety signal.

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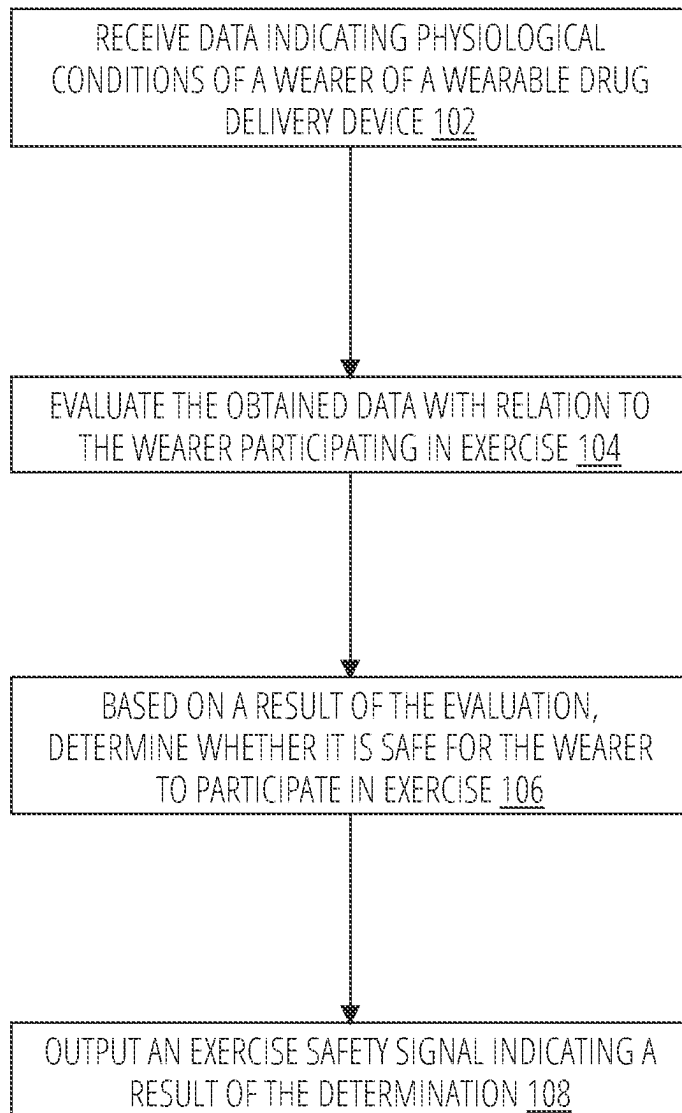
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**FIG. 1**

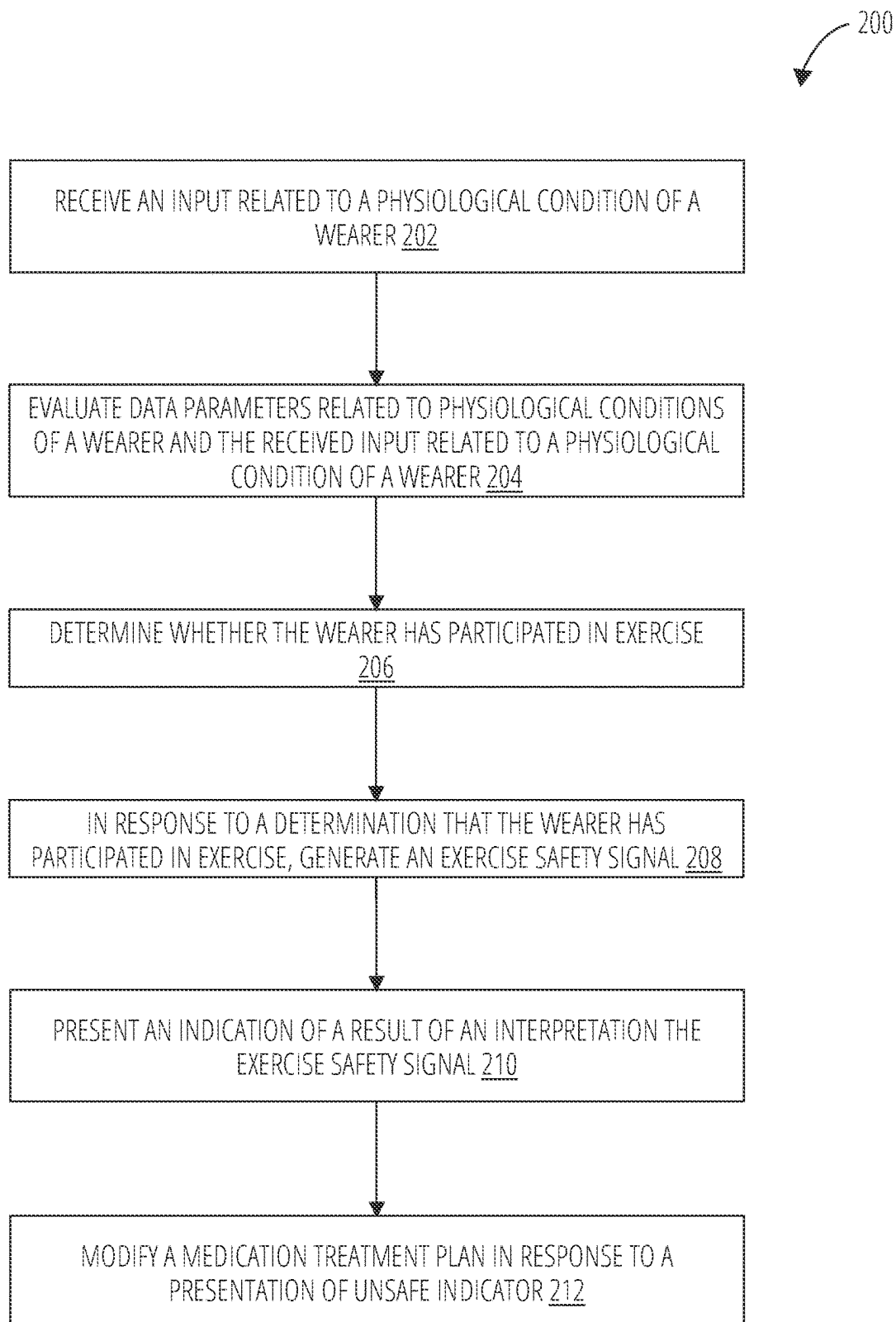


FIG. 2

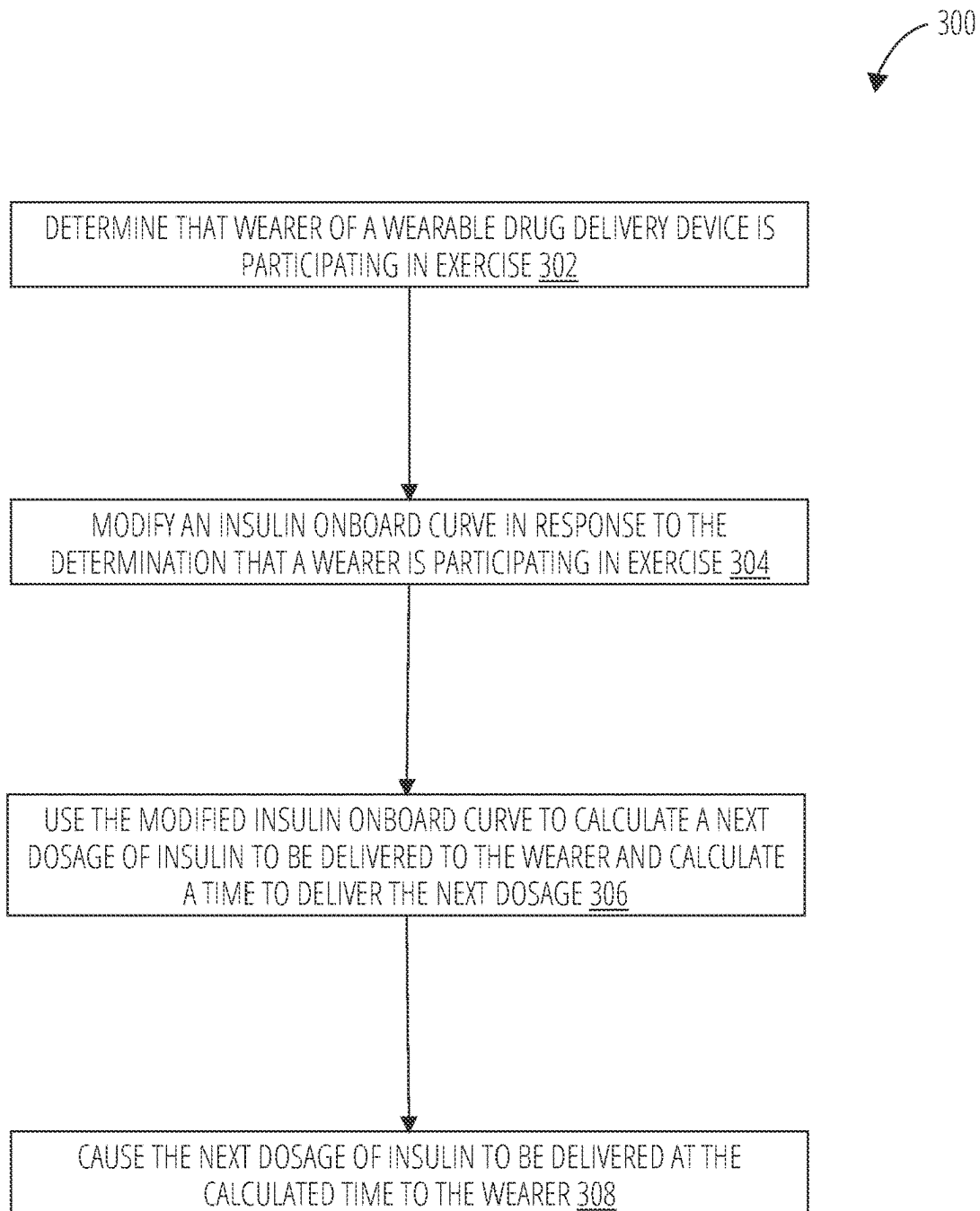


FIG. 3

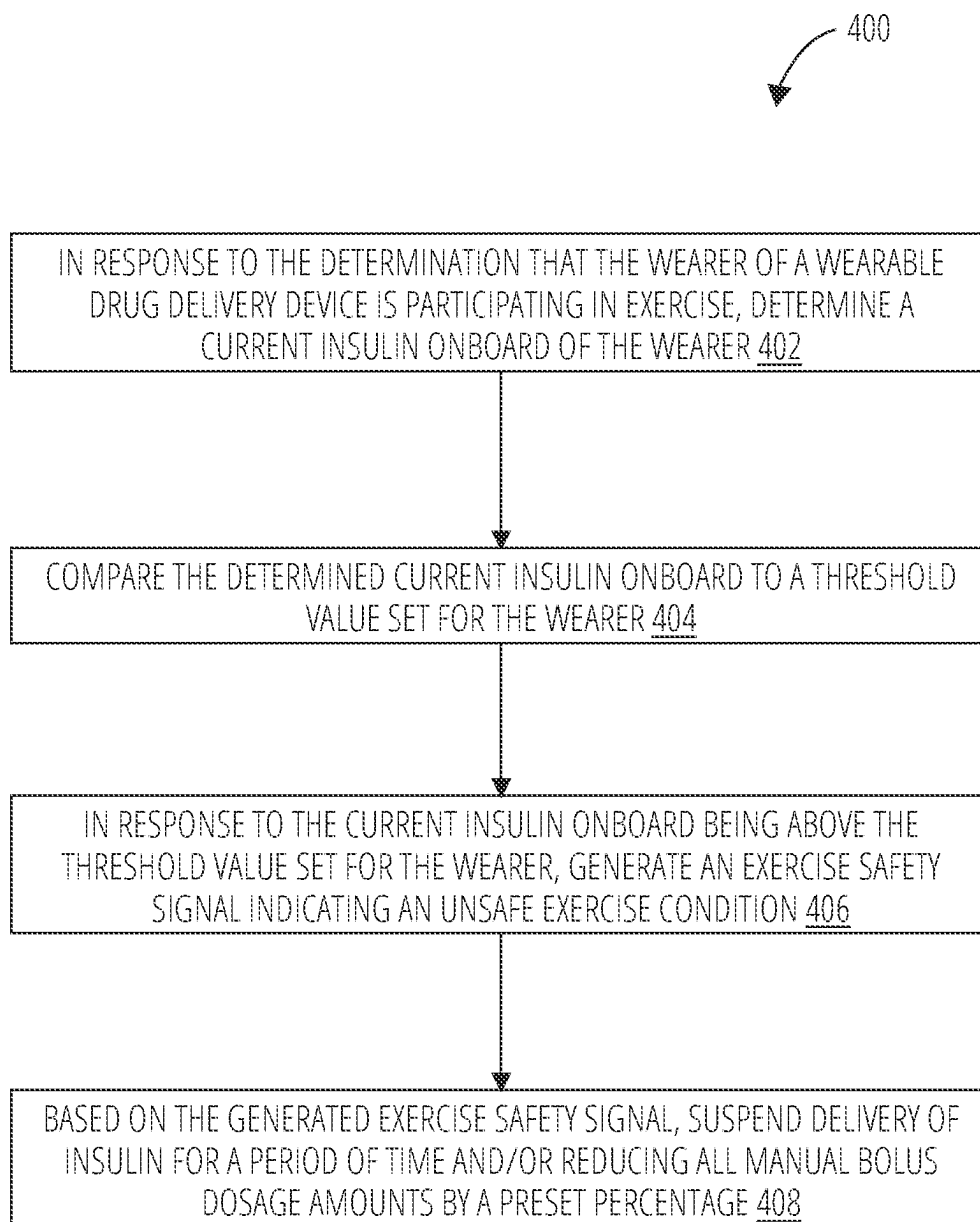


FIG. 4

500 →

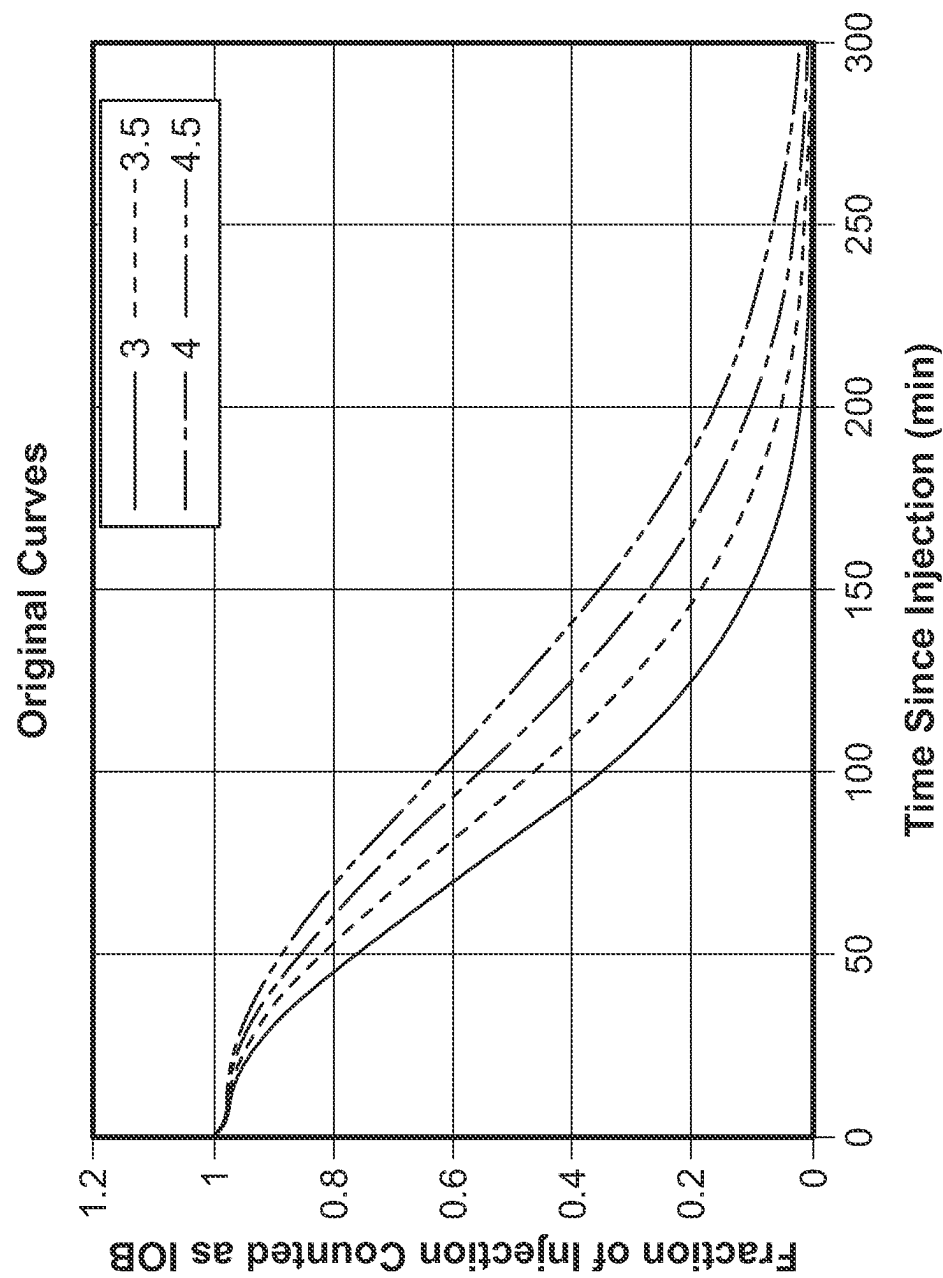


FIG. 5

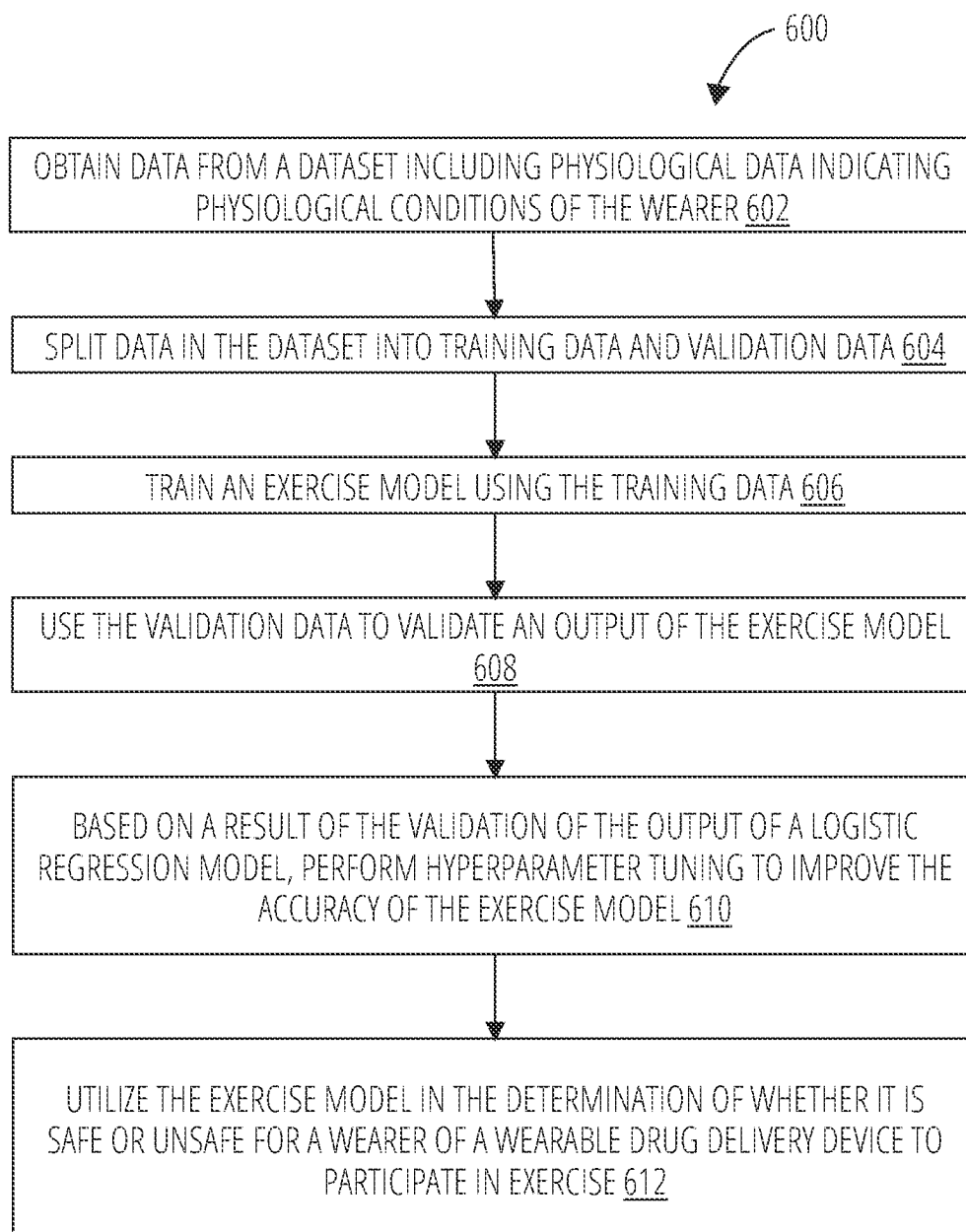


FIG. 6

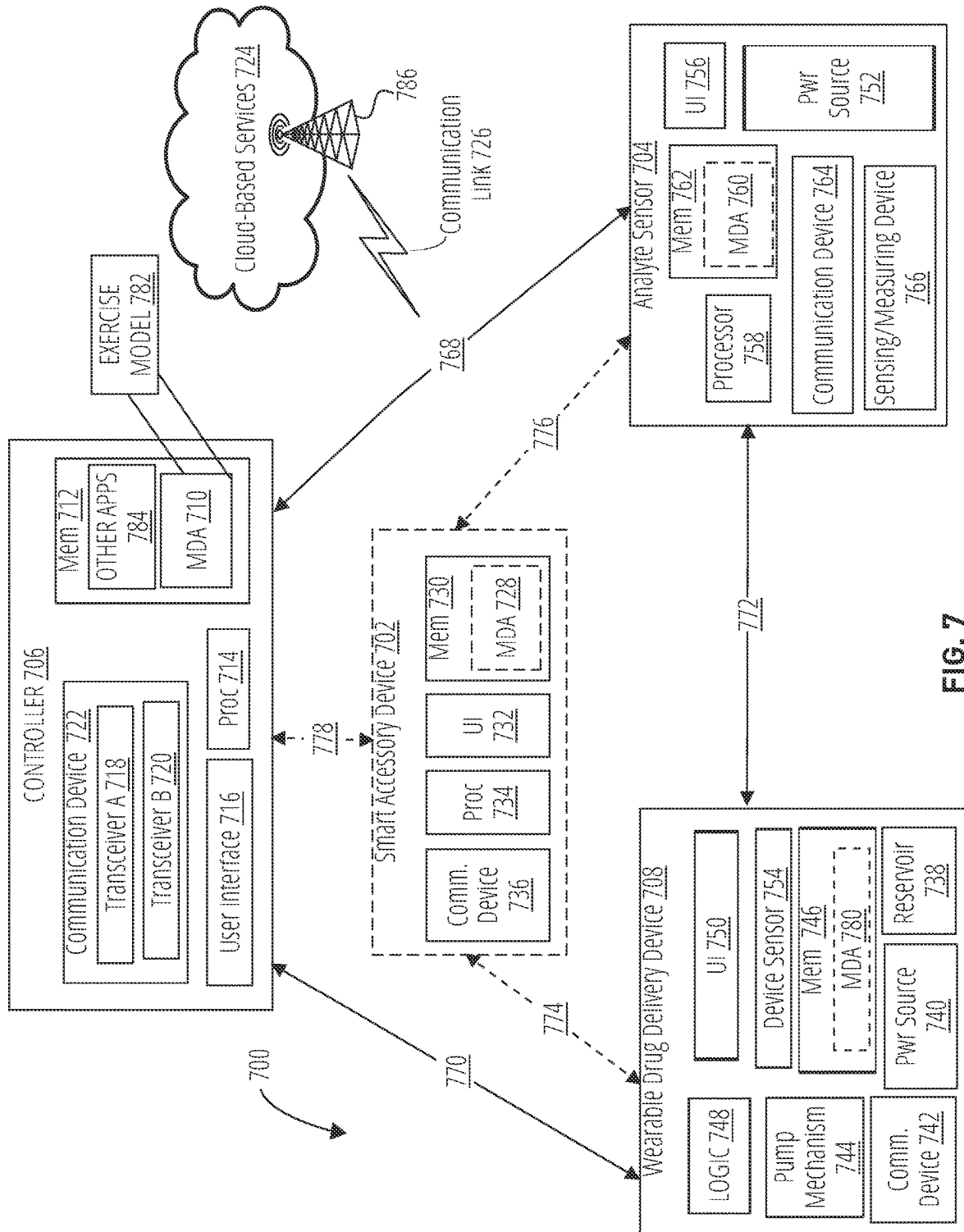


FIG. 7

1

EXERCISE SAFETY PREDICTION BASED ON PHYSIOLOGICAL CONDITIONS

RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Patent Application No. 63/197,613, filed Jun. 7, 2021, the contents of which are incorporated herein by reference in their entirety.

BACKGROUND

Maintaining healthy lifestyle is both critically important to patients with Type 1 diabetes and challenging at the same time. Many patients have a challenge to maintain their blood glucose level at an appropriate level during or after exercise and may have to take rescue snacks if there is a concern of hypoglycemia. Exercise increases insulin sensitivity and muscle cells are better able to use any available insulin to take up glucose during and after activity. Moreover, during exercise, muscle cells absorb glucose for energy, whether insulin is available or not. As a result, glucose in the blood decreases, leading to potential hypoglycemic events, and causing diabetics to feel fearful of exercise and hypoglycemia due to the over delivery of insulin during or after exercise.

These situations can become more difficult to manage if the patient has a high amount of insulin onboard (JOB) or just had either a post-prandial bolus or a correction bolus. The excess insulin followed by exercise activity may further increase the risk of hypoglycemia as insulin absorption increases from exercise, the exercise itself causes the body to metabolize glucose more quickly, and the additional insulin further acts to reduce blood glucose.

BRIEF SUMMARY

In one aspect, a wearable drug delivery device includes a processor, a pump mechanism, a memory, and a reservoir. The reservoir may contain a liquid drug. The pump mechanism may be communicatively coupled to the processor and fluidically coupled to the reservoir. The memory may be coupled to the processor and operable to store programming code that, when executed by the processor, causes the processor to be operable to receive data indicating physiological conditions of a wearer of the wearable drug delivery device, and evaluate the obtained data with relation to the wearer participating in exercise. The processor may determine whether it is safe for the wearer to participate in exercise based on a result of the evaluation and output an exercise safety signal indicating a result of the determination of whether it is safe or unsafe for the wearer to participate in exercise.

In another aspect, an automatic, wearable drug delivery system may include a drug delivery device and a controller. The drug delivery device may include logic circuitry operable to control the drug delivery device, a reservoir configured to contain a liquid drug, a communication device coupled to the logic circuitry and a pump mechanism responsive to the logic circuitry and fluidically coupled to the reservoir. The controller may include a processor and a memory. The memory may be coupled to the processor and configured to store a medication delivery application, an exercise safety application, and programming code. When the exercise safety application is executed by the processor, the processor is operable to receive an input from the wearer related to a physiological condition of a wearer. The pro-

2

cessor may evaluate data parameters related to physiological conditions of a wearer and the received input related to the physiological condition of a wearer and determine whether the wearer has participated in exercise. The processor, in response to a determination that the wearer has participated in exercise and a result of the evaluation of the data parameters, may generate an exercise safety signal, and present an indication of a result of an interpretation of the exercise safety signal at a user interface. The indication is a safe or unsafe indicator. In response to the generation of the exercise safety signal, the processor may modify a medication treatment plan.

In a further aspect, a non-transitory computer-readable storage medium is provided that may include instructions that, when executed by a processor, cause the processor to determine that a wearer of a wearable drug delivery device is participating in exercise. The instructions may cause the processor to modify an insulin onboard curve in response to the determination that a wearer is participating in exercise and use the modified insulin onboard curve to calculate a next dosage of insulin to be delivered to the wearer and calculate a time to deliver the next dosage. The instructions may trigger the processor to cause the next dosage of insulin to be delivered at the calculated time to the wearer.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a flowchart of an example process implemented according to an aspect of the disclosed subject matter.

FIG. 2 is a flowchart of another example process implemented according to a further aspect of the disclosed subject matter.

FIG. 3 is a flowchart of a further example process implemented according to another aspect of the disclosed subject matter.

FIG. 4 illustrates in a flowchart a process in accordance with an embodiment of the disclosed subject matter.

FIG. 5 is a modified insulin onboard decay curve that illustrates an example usable in an embodiment of the disclosed subject matter.

FIG. 6 illustrates an example of a process for generating an exercise model usable with the aspects of the disclosed subject matter.

FIG. 7 illustrates a functional block diagram of a system example suitable for implementing the example processes and techniques described herein.

DETAILED DESCRIPTION

Systems, devices, computer-readable medium and methods in accordance with the present disclosure are now described more fully hereinafter with reference to the accompanying drawings, where one or more embodiments are shown. The systems, devices, and methods may be embodied in many different forms and are not to be construed as being limited to the embodiments set forth herein. Instead, these embodiments are provided so the disclosure is thorough and complete, and may fully convey the scope of methods and devices to those skilled in the art. Each of the systems, devices, and methods disclosed herein provides one or more advantages over conventional systems, components, and methods.

Various examples provide a method, a system, a device and a computer-readable medium for responding to inputs provided by sensors, such as an analyte sensor, and wearers (i.e., users) of an automatic drug delivery system. "Analyte sensor" may refer to a wearable device that includes sensing

and measurement devices configured to detect different analytes in a wearer's body and particularly in the user's blood, such as blood glucose levels, and/or other analytes as described herein. The various devices and sensors that may be used to implement some specific examples may also be used to implement different therapeutic regimens using different drugs than those described in the specific examples.

The disclosed examples provide techniques related to the safety of a wearer to participate in exercise that may be used with any additional algorithms or computer applications that manage blood glucose levels and insulin therapy. These algorithms and computer applications may be collectively referred to as "medication delivery algorithms" or "medication delivery applications" and may be operable to deliver different categories of drugs (or medications), such as chemotherapy drugs, pain relief drugs, diabetes treatment drugs (e.g., insulin, glucagon and/or glucagon-like peptides), blood pressure medication, or the like.

A type of medication delivery algorithm (MDA) may include an "artificial pancreas" algorithm-based system, or more generally, an artificial pancreas (AP) application. For ease of discussion, the computer programs and computer applications that implement the medication delivery algorithms or applications may be referred to herein as an "AP application." An AP application may be configured to provide automatic delivery of insulin based on an analyte sensor input, such as signals received from an analyte sensor, such as a continuous blood glucose monitor, or the like. The signals from the analyte sensor may be blood glucose measurement values, or the like.

In one aspect, a wearable drug delivery device may be operable to determine whether it is safe for a wearer of the drug delivery device to participate in exercise at a present time or future time. The wearable drug delivery device may include, for example, a processor, a reservoir, a pump mechanism, a communication device, and a memory. The reservoir may be configured to contain a liquid drug. The pump mechanism may be communicatively coupled to the processor and fluidically coupled to the reservoir. The memory may be coupled to the processor and configured to store programming code that, when executed by the processor, is operable to perform different functions. The wearable drug delivery device may be communicatively coupled to a controller device. The controller device may have a processor and memory as well as communication devices that enable communication with the wearable drug delivery device and also external devices and systems, such as cloud-based services. FIG. 1 is a flowchart of an example process implemented according to an aspect of the disclosed subject matter.

The process 100 may be executed by the processor of the controller as the controller processor may have greater computational power than a processor of the wearable drug delivery device. In addition, the controller memory may maintain a greater amount of historical data than the memory of the wearable drug delivery device.

In block 102, the processor when executing the programming code that implements process 100 may receive data that indicates physiological conditions of a wearer of the wearable drug delivery device. The received data may include a current blood glucose measurement value or data indicative thereof, a total daily insulin value, an amount of a requested bolus, a time at which the bolus was requested, a current insulin onboard value, or an amount of insulin delivered within a past evaluation period. A past evaluation period may be a previous cycle, such as 5 minutes, or another period of time such as a past 15-minute period, a

30-minute period, a 60-minute period, a 120 minute period of time, or a set time period such as when a wearer may most likely exercise, such as between 5 am and 7 am, 12 pm to 1 pm, or the like.

In addition, or alternatively, the processor, when executing the stored programming code, may be further operable to retrieve physiological data from the memory, where the physiological data may contain a number of days of data related to the wearer's participation in exercise. The processor may use the received data, the retrieved physiological data or both to update an exercise model.

In block 104, the process 100 may evaluate the obtained data with relation to the wearer participating in exercise. For example, the processor, when evaluating the obtained data with relation to the wearer participating in exercise, may be operable to input values obtained from the received data into an exercise model (which may be the updated exercise model), and analyze an output of the exercise model with reference to threshold values related to exercise safety for the wearer. The threshold values related to exercise safety may be parameters that are weighted differently for each wearer according to wearer demographics, disease progression (e.g., type 2 diabetes versus type 1 diabetes), or the like.

In block 106, the processor implementing process 100 may, based on a result of the evaluation, determine whether it is safe for the wearer to participate in exercise. For example, the processor may be further operable to determine whether the result of the evaluation indicates a presence of one or more conditions and based on the determination of the presence of the one or more conditions. For example, the output of the exercise model may indicate whether it is safe or unsafe for the wearer to exercise and the processor may use the output from the exercise model to make the determination.

In addition, or alternatively, the processor, when determining whether it is safe for the wearer to participate in exercise, may be further operable to determine whether the result of the evaluation indicates that a trajectory of blood glucose measurement values of the wearer is trending downward. Based on the determination that the trajectory of the glucose measurement values of the wearer are trending downward, the processor may generate an indication that exercise is unsafe. For example, the downward trend may be a rate of change of $(-)$ 100 mg/dL per hour for the trajectory of blood glucose measurement values of the wearer. A downward trend may also be referred to as a negative rate of change or a negative (or downward) trajectory of blood glucose measurement values. Of course, rates of change other than $(-)$ 100 mg/dL may be used, such as $(-)$ 10 mg/dL, $(-)$ 50 mg/dL, $(-)$ 75 mg/dL, $(-)$ 90 mg/dL, or the like. Additionally, or alternatively, the output of the exercise model may indicate whether it is safe or unsafe for the wearer to exercise based on a current or future predicted value of the user's blood glucose level. For example, the exercise model may make a prediction based on current blood glucose trends where the user's blood glucose measurement value will be in 30 minutes or 1 hour, and if this predicted value is less than X mg/dL (e.g., where X is approximately 70, 80, 90, 100, or 110 mg/dL), the processor may output an indication that it is unsafe to exercise, or an indication that it is unsafe to exercise for the next Y minutes (for example, where Y is approximately 30 or 60 minutes), or an indication that it is unsafe to exercise unless the user consumes food or carbohydrates, or an indication that it is safe to exercise, or a combination of these potential outputs.

In block 108, process 100 may cause output of an exercise safety signal indicating a result of the determination of

5

whether it is safe for the wearer to participate in exercise. Based on the determination of whether it safe or unsafe for the wearer to participate in exercise, the processor may generate an indication that exercise is unsafe. Alternatively, the processor may generate that exercise is safe. For example, as described in more detail with reference to other figures, the wearable drug delivery device may also include an output device coupled to the processor. The processor may be further operable to interpret the outputted exercise safety signal. Based on the interpretation, the processor may provide an output signal to the output device based on the outputted exercise safety signal.

For example, the processor may interpret the outputted exercise safety signal as indicating that it is unsafe for the user to exercise. The processor may output command signals that cause the output device to make outputs that are indications of unsafe exercise situations, such as vibrating in a particular pattern of vibrations (e.g., Morse code for distress—S-O-S), make a sound (e.g., a particular beeping sound or number of beeps), change color (e.g., turn from “green” to “yellow” or “red” depending upon the level of urgency (i.e., the degree of unsafety)), or the like. Alternatively, in response to the processor interpreting the outputted exercise safety signal as indicating it is safe to exercise, the processor may output command signals that cause the output device to make outputs that are indications of safe exercise situations, such as vibrating in a particular pattern of vibrations (e.g., 2 vibrations, or the like), make a sound (e.g., a different beeping sound or number of beeps as compared with an unsafe signal), change color (e.g., turn from no lights to a “green” light or from “yellow” to “green”), or the like. Alternatively, if the processor interprets the exercise safety signal as indicating it is safe to exercise, the processor may not generate any output signals.

In addition, or alternatively, the processor may be further operable to modify a medication treatment plan based on the outputted exercise safety signal. For example, if the processor outputs an exercise safety signal indicating it is unsafe to exercise, but the wearer is determined to participate in exercise, the processor may cause all delivery of insulin to cease for a period of time or reduce an amount of insulin that may be delivered to the user for period of time, included in manual bolus dosages. The cessation or reduction may vary from wearer to wearer.

The determination that the wearer is participating in exercise may be made according to various techniques. For example, the wearer may self-report the participation in exercise by inputting exercise participation in a graphical user interface (provided via an exercise detection application or the AP application) presented on a display of a controller. Alternatively, the controller and/or wearable drug delivery device may have sensors, such as an accelerometer, global positioning system (GPS) signals, Wi-Fi signals, gyroscopes, or the like, that output signals in response to movement by the wearer, a change in location, or an indication of location. Using the information provided by one or more of the respective sensors, the processor when evaluating the information may determine that the information has previously indicated participation in exercise or a location where exercise takes place (e.g., a gym, a ballpark, soccer field, or the like). In addition, or alternatively, the processor may execute an exercise detection application or algorithm that is operable to evaluate inputs from the sensors as well as location information based on the GPS or Wi-Fi signals, information obtained from other mobile computer applications, such as fitness applications, calendar applications (e.g., fitness class schedules, athletic team schedules, or the

6

like). Based on the evaluated inputs, such as high acceleration values for an extended period of time (duration threshold) or a certain number of high acceleration values within another set threshold period of time, a calendar entry indicating the current time as a scheduled time of a fitness class, the processor may determine above a preset probability that the wearer is participating in exercise.

Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

In another aspect, a wearable drug delivery system may include a drug delivery device, an analyte sensor, and a controller as well as other devices described with reference to other examples. FIG. 2 is a flowchart of another example process implemented according to a further aspect of the disclosed subject matter. The process 200 may be executed by a processor such as the processor of a controller (shown in other examples). For example, the controller may be a smartphone or a dedicated mobile computing device. The processor of the controller may be operable to execute many computer applications including a calendar application, a fitness application, a medication delivery application that utilizes an exercise safety algorithm, such as process 200, and the like.

In block 202, the processor may receive an input related to a physiological condition of a wearer. The input may, for example, be from the wearer or from an exercise safety algorithm. In an example, the wearer may request a bolus, use a hypoglycemia protect mode, or indicate that they are about to participate in exercise, or schedule a time at which they will participate in exercise.

In block 204, in response to the received input related to a physiological condition of a wearer, the process 200 may evaluate data parameters related to one or more physiological conditions of a wearer. The data parameters related to physiological conditions of a wearer may, for example, include a blood glucose measurement value, a total daily insulin amount, a current calculation of an amount of insulin onboard, and an amount of insulin delivered in a past evaluation period.

For example, the processor, when evaluating the data parameters, may be further operable to input data parameters including a blood glucose measurement value, a total daily insulin value, a time and an amount of a requested bolus, an amount of insulin onboard value, or an amount of insulin delivered within a past evaluation period, into an exercise model. The result of the evaluation of the data parameters at block 204 may be an indication that exercise is unsafe or safe at the time of the evaluation. For example, the data parameters may be evaluated utilizing a machine-learning, exercise model that outputs a value indicating whether participation in exercise by a user or wearer is safe or unsafe.

In block 206, the processor determines whether the wearer has participated in exercise. Participation in exercise may be determined as mentioned above with reference to the exercise determination in FIG. 1. In addition, or alternatively, the determination of exercise participation may be made based on an analysis of different attributes and factors, such as the user's calendar application on the controller, an exercise input entered through a user interface of the controller, an exercise detection application executing on the controller, based on outputs from other sensors on the controller, such as an accelerometer, gyroscope, camera or microphone, or the like. The factors or attributes may be inputs into an exercise model that is operable to determine whether exercise is safe or unsafe based on physiological conditions of a wearer.

In block 208, in response to a determination that the wearer has participated, or is participating, or will participate in exercise, the processor may generate an exercise safety signal. For example, the processor may use the result of the evaluation of the data parameters obtained in block 204 when generating the exercise safety signal. For example, after participating in exercise, the wearer may indicate that they are feeling “dizzy” or “lightheaded,” and, as a result, the processor can identify (e.g., label) physiological data (e.g., data parameters such as blood glucose measurement values, a total daily insulin amount, a current calculation of an amount of insulin onboard, and an amount of insulin delivered in a past evaluation period) that correspond to the indicated feeling of dizziness or lightheadedness.

Additionally, the processor may be operable to control a communication device to establish a wireless communication connection with an external device, where the external device is a data network device (shown in another example) or a cloud-based service (also shown in another example) to access external devices or services, and retrieve physiological data from a memory source (such as cloud-based storage or the like), where the physiological data contains a number of days of data related to the wearer’s participation in exercise. The wearable drug delivery device may also include updates to an exercise model using the retrieved physiological data.

In block 210, process 200 when executed may cause the processor to present an indication of a result of an interpretation of the exercise safety signal, wherein the indication is a safe or unsafe indicator for exercise participation. For example, the presentation of the safe or unsafe indicator may be on a display of a controller, such as a touchscreen display, or a user interface of a drug delivery device, or the like.

Alternatively, in a further embodiment, based upon the construction of the exercise model, personalized data and daily blood glucose measurement value trend averages, the processor may suggest the best suitable times for the wearer to participate in exercise. As such, the presented indication may suggest to the wearer to wait 15 minutes, 30 minutes or the like before participating, suggest alternative exercises to perform (e.g., weightlifting (short bursts) versus a long run (an extended high heart rate), or recommend ingesting carbohydrates (e.g., consuming a snack or the like) before exercising.

In addition to predicting safe or unsafe conditions, the exercise model may be further trained to predict a class of probabilities that may be used as threshold values related to exercise safety. In this case, the model may predict which intensity of the exercise may be unsafe. For example, for given physiological conditions of a wearer, high intensity training may not be suitable at this time for the wearer, but a brisk walk or yoga might be acceptable. The recommen-

dations for the type of exercise to participate in and/or to avoid may be presented to the wearer via a user interface of a controller or other device. In addition, or alternatively, a class of probabilities may be selected such that the model favors false positives over false negatives because the consequences of false negatives are higher. This is similar to screening tests where further investigation is needed. In this example, the model may favor a conservative approach where a signal indicating that it is unsafe to exercise is better than the converse. The probabilities could be constructed by retrospect of past glucose distributions under the same activities. For example, if a current blood glucose measurement value is below the 10th percentile of the blood glucose measurement values during past training activities, then the processor executing an exercise safety algorithm may determine that it is not safe for the wearer to do the same activity as in the past training activities at this time.

In block 212, the processor when executing process 200 may cause a medication delivery application to modify a medication treatment plan in response to a presentation of unsafe indicator. For example, if the processor determines it is unsafe to exercise the processor may withhold delivery of insulin for a period of time. Alternatively, if the processor determines it is safe to proceed with exercise, the processor may cause a modification of a dosage of insulin but continue with the presently set delivery schedule for the next dose.

Either of the steps in block 210 or 212 may be optional depending upon user preference settings or clinician settings or other settings (e.g., default settings). For example, a user preference may be set to notify the user only when exercise is unsafe and no notification when exercise is safe. Other user preference settings may be, for example, to receive notifications whether exercise is safe or unsafe, or receive a notification only when exercise is determined to be safe. It may be helpful to have a couple of exemplary physiological conditions of a wearer as well as other factors that are evaluated by the exercise model.

Each row of Table 1 below may represent a different day for a particular wearer or a group of different wearers. In the table below, the exercise model may evaluate the physiological data related to the physiological conditions of a wearer, such as one or more blood glucose measurement values provided by a continuous glucose monitor (CGM), an indication of whether any blood glucose measurement values were missed, a number of boluses delivered, an amount of insulin (in Units) delivered within the last 30 minutes, whether the wearer’s blood glucose measurement value was within the hypoglycemic range (for the wearer, such as, for example, 60 mg/dL or lower), a calculated amount of insulin onboard (IOB) for the wearer, the wearer’s target blood glucose setting, and the wearer’s TDI. The label is the binary indication of safe to exercise (Label=0) or unsafe to exercise (Label=1).

TABLE 1

CGM	Missed count	bolus	Insulin in last 30 min	intensity of exercise	IOB	setpoint	TDI	label
160	0	0	0	1	0.421	110	45	0
159	0	0	4	1	0.374	110	100	0
158	1	0	10	2	4.332	110	21	1
153	0	2	0	1	0.257	110	25	0
152	1	0	0	1	0.225	110	32	0
151	1	0	4	1	0.196	110	5	0
206	5	0	5	1	0.456	110	40	0
200	0	0	6	2	1.444	110	10	1
192	0	0	0	0	0.430	110	30	0

TABLE 1-continued

CGM	Missed count	bolus	Insulin in last 30 min	intensity of exercise	IOB	setpoint	TDI	label
149	0	0	0	0	0.321	110	65	0
80	0	5	3	0	1.301	110	40	1
132	0	0	0	0	0.280	110	21	0
126	0	0	0	0	0.260	110	21	0
72	0	2	10	0	3.073	110	50	1
71	0	0	2	0	0.064	110	21	0
-1	18	4	5	2	1.031	110	90	1
70	1	0	2	0	0.026	110	100	0

The “intensity of the exercise” data point indicates how intense the exercise is to the user, for example, intensity values may be: 1=normal (E.g. normal walk, etc.), 2=moderate (brisk walk, biking, etc.), and 3=intense (high intensity interval training (HIIT), running, etc.)

In an example using the third row of Table 1 above, the exercise model may indicate that the physiological data in the third row triggers an unsafe exercise response as indicated by the label=1. The basis for the exercise model indicating that the physiological data indicates may be based for example on a low TDI of 21 Units (U), the 10 U insulin in past 30 minutes, and a current IOB of approximately 4.332 Units. Together, these parameters, when combined with the user’s performance of intense exercise, or perhaps even mild exercise (given the particular user of a wearable drug delivery device), and when evaluated by the exercise model, may indicate that participation in exercise by the user is either safe or unsafe.

The other rows of data that resulted in being labeled unsafe were based on various factors. For example, the row in which the CGM value is 200, the data that generated the unsafe to exercise indication was based on the low TDI (10), the 6 U of insulin in the past 30 mins, current IOB of approximately 1.444 U. Similarly, the third indication of unsafe to participate in exercise is in the CGM reading of 80, the data that generated the unsafe to exercise indication was based on a low CGM reading (80), a medium range TDI of approximately 40, the medium range IOB of approximately 1.301, and the 5 U requested bolus. The fourth indication of unsafe to participate in exercise is in the CGM reading of 72, the data that generated the unsafe to exercise indication was based on a low CGM reading (72), a medium range TDI of approximately 50, the higher IOB of approximately 3.703, the 10 U of insulin in the last 30 minutes, and the 2 U requested bolus. The fifth indication of unsafe to participate in exercise is in the CGM reading of -1, which suggests that CGM readings have been missed past a threshold amount (for example, for the past 30 minutes), and in this case, for past 90 minutes (5 minutes times 18 missed readings), the medium range of IOB of approximately 1.031, the 5 U of insulin in past 30 mins, a requested bolus of 4 U and the TDI of 90.

Based on the label, the programming code may cause the processor to generate an exercise safety signal indicating whether exercise is safe or unsafe. The processor may respond to the generated exercise safety signal as discussed above. The exercise model may be a machine learning model, such as a linear regression model, a logistic regression model, a decision tree model, a support vector machine, a naïve Bayes classifier, or the like.

It may be helpful to describe at a high level, details related to the exercise model, which in the example below is a logistic regression model, and, in particular, how the exercise model may be built, trained, and modified, if necessary.

The exercise model may be generated by a processor in a server, such as a cloud-based server, or the like.

In an operational example of how the logistic regression exercise model is built; a processor may be further configured to obtain data from a dataset including physiological data indicating physiological conditions of the wearer. The processor may split data in the dataset into training data and validation data. Using the training data, a logistic regression model may be trained. The validation data may be used to validate an output of the logistic regression model. Based on a result of the validation of the output of the logistic regression model, the processor may perform hyperparameter tuning to improve the accuracy of the logistic regression model. The processor may retrieve the received data related to physiological conditions of the wearer and input the retrieved current data into the logistic regression model for evaluation. The processor may, for example, label the output data as either indicating an unsafe condition for exercise or a safe condition for exercise based on a value of output data from the logistic regression model. The processor may generate an exercise safety signal based on any label indicating an unsafe condition for exercise.

Once you build a classifier and enable a Boolean output, the classifier may provide a probability related to whether it is safe to exercise. A threshold based on past history (whether it was safe to exercise in the past) may be established, and everything above threshold would be true, everything below would be false. Boolean value of true or 1 would be unsafe to exercise.

Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

FIG. 3 is a flowchart of a further example process implemented according to another aspect of the disclosed subject matter.

A processor or logic circuitry may be operable to execute programming code stored in a memory coupled to the respective processor or logic circuitry. In block 302, a processor executing process 300 may be operable to determine that a wearer of a wearable drug delivery device is participating in exercise. It is noted that exercise can have both short- and long-term physiological effects.

Once participation in exercise is detected, the system can implement a long-term lengthening of the IOB curves over a period of 12-24 hours following exercise to account for the long-term effects of exercise. For example, as shown in block 304, the process 300 modifies an insulin onboard curve in response to the determination that a wearer is participating in exercise. The processor when executing process 300 may modify the insulin onboard (IOB) curve by extending a length of the insulin onboard curve. Extending the length of the IOB curve slows decay of a calculated amount of insulin onboard and the extension remains in place for a preset period of time.

11

The modification of the IOB curve makes the processor evaluate the effect of exercise based on changing the effective IOB (i.e., the amount of insulin onboard that will be metabolized by the wearer's body over time), while the wearer's sensitivity to insulin remains unchanged because a change to sensitivity would be a direct change to the exercise model. Instead, the modification of the length of the IOB curve makes a change to the perceived length of time the insulin is acting in the wearer's body. Based on the extended IOB curve, the processor may modify constraints that are applied to the insulin doses (e.g., limits on the maximum amount of insulin that is included in a bolus, delivered within X number of hours of the determined participation in exercise, or the like, where X is approximately 3.5 to approximately 4.5) but the medication delivery algorithm continues making insulin delivery decisions based on threshold values related to exercise safety and the modified constraints. For instance, the assumed 3-hour and 4-hour IOB decay curves may be lengthened to 3.5 hours and 4.5 hours following detection of significant exercise, to incorporate the impact of relatively longer-lasting increased insulin sensitivity. Of course, other values of X may be 1.5, 2, 6, 12 or the like. This is illustrated in the example IOB curves of FIG. 5.

In block 306, the processor implementing process 300 may use the modified insulin onboard curve to calculate a next dosage of insulin to be delivered to the wearer and calculate a time to deliver the next dosage.

In block 308, the process 300 may cause the next dosage of insulin to be delivered at the calculated time to the wearer.

Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

FIG. 4 illustrates a process 400 in accordance with an embodiment of the disclosed subject matter.

In block 402, a processor executing process 400 may be operable to, in response to the determination that the wearer of a wearable drug delivery device is participating in exercise, determine a current insulin onboard of the wearer.

In block 404, process 400 compares the determined current insulin onboard (IOB) to a threshold value set for the wearer. The threshold value for insulin onboard may be one of the threshold values related to exercise safety that is evaluated by the processor. In this example, the current IOB may be a primary factor used in the determination of whether participation in exercise is safe or unsafe for the wearer of the wearable drug delivery device. For example, an IOB of 3 U or greater may indicate that the artificial pancreas application is overcompensating or overdelivering insulin, and hence it is unsafe for the user to exercise. Further, the processor may also determine whether the wearer's total daily insulin (TDI) is going to reach a threshold TDI value, such as, for example, 4 times, 5 times, or 9 times the expected total daily insulin delivery in a 3-hour period of time or another period of time, such as 2-hour period of time, and hence make a determination that it is unsafe to exercise. Alternatively, another threshold may be one in which the total insulin delivered over the last 3 hours, including manual boluses and system automated deliveries, in not permitted to exceed 15 times the number of basal hours, or 31% of TDI. Of course, other TDI thresholds that indicate overcompensation or over delivery may be used.

In block 406, process 400 in response to the current insulin onboard being above the threshold value set for the wearer, generates an exercise safety signal indicating an unsafe exercise condition. In block 408, process 400 based on the generated exercise safety signal, suspends delivery of

12

insulin for a period of time and/or reducing all manual bolus dosage amounts by a preset percentage, such as 25%, 50%, 75%, or in some cases 100%.

FIG. 5 is a modified insulin onboard decay curve that illustrates an example usable in an embodiment of the disclosed subject matter.

In modified insulin onboard decay curve 500 of FIG. 5, the solid line represents the original 3-hour curve, the dashed line represents the 3-hour insulin decay curve that has been extended by 30 minutes (i.e., the 3.5-hour insulin delay curve), the 4-hour insulin delay curve is shown as dash-dot-dash, and the dashed-dot-dot-dashed line represents the respective 4-hour insulin decay curve extended by 30 minutes (i.e., the 4.5-hour insulin delay curve). As can be seen in the curves, for example, at the 100th minute mark, the remaining insulin action is approximately 36% with the 3-hour insulin delay curve, but approximately 46% with the 3.5-hour insulin delay curve, resulting in increased estimated remaining insulin on board and thus potentially a more conservative insulin delivery by the algorithm. If IOB is higher, the insulin delivery algorithm will tend to deliver less insulin.

The modifications of the IOB curves may vary based on the system's original IOB curves that are utilized. The original IOB curves may be based on anonymized, clinical data gathered from hundreds or thousands of patients or based on a specific wearer's history.

FIG. 6 illustrates an example of a process for generating an exercise model usable with the aspects of the disclosed subject matter. The generation of the exercise model may occur in a server, such as a server provided by cloud-based services or the like, that is also communicatively coupled to the controller as described with reference to a later example.

As previously mentioned, the exercise model may be a machine learning model that is capable of modifying its output based on the values input into the model and a closeness of the output value to a result produced from a validation data set. In the following discussion of routine 600, the exercise model is implemented as a logistic regression model.

In block 602, a server executing routine 600 may obtain data from a dataset that includes physiological data indicating physiological conditions of the wearer. For example, the controller may be operable to upload anonymized physiological data indicating the physiological conditions of the wearer to the cloud-based services. The data uploaded from the controller may be combined with similar anonymized data that is uploaded from other controllers. Alternatively, the controller may be operable to obtain and process the data. For example, the controller may obtain the data from a memory coupled to the controller or from a server or data storage accessible through cloud-based services. The obtained data may be an entire data set or part of a data set, such as a past evaluation period which may be three, six, nine days, or the like.

In block 604, the server or the controller may split data in the dataset into training data and validation data.

In block 606, a logistic regression model may be trained by the server or the controller using the training data.

In block 608, the server or the controller may use the validation data to validate an output of the logistic regression model.

Based on a result of the validation of the output of the logistic regression model, the controller or server may in block 610 perform hyperparameter tuning to improve the accuracy of the logistic regression model. Once the exercise model is appropriately trained, the controller may begin

13

executing the exercise model to determine whether it is safe or unsafe for the wearer to participate in exercise based on the wearer's physiological conditions.

For example, the controller may in block 612 be operable to utilize the exercise model in the determination of whether it is safe or unsafe for a wearer of a wearable drug delivery device to participate in exercise.

In the examples described with reference to FIGS. 1-6, a controller may input data related to the physiological conditions of the wearer into the exercise model and obtain an exercise safety signal. The data related to the physiological condition of the wearer may be blood glucose measurement values, IOB, total daily insulin (TDI), and similar data as described above with reference to other examples.

FIG. 7 illustrates a functional block diagram of a system example suitable for implementing the example processes and techniques described herein.

The automatic wearable drug delivery system 700 may implement (and/or provide functionality for) a medication delivery algorithm (MDA), such as an artificial pancreas (AP) application, to govern or control automated delivery of a drug, a therapeutic, or a medication, such as insulin, to a user (e.g., to maintain euglycemia—a normal level of glucose in the blood). The automatic wearable drug delivery system 700 may, for example, include an analyte sensor 704, a controller 706, a wearable drug delivery device 708, and an optional smart accessory device 702.

The controller 706 may be remote from the wearable drug delivery device 708 and may include a user interface 716, a communication device 722, a memory 712, and a processor 714. The user interface 716 is coupled to the processor 714 and operable to receive inputs related to a physiological condition of a wearer and provide the input to the processor 714. In an example, the input may be a request for a bolus dosage. The controller 706 may include a user interface 716, which may be a keypad, a touchscreen display, levers, light-emitting diodes, buttons on the controller 706, a microphone, a camera, a speaker, a display, or the like, that is configured to allow a user to enter information and allow the controller 706 to output information for presentation to the user (e.g., alarm signals, exercise recommendations (e.g., exercise times and/or exercise intensity, and the like). The user interface 716 may provide inputs, such as a voice input, a gesture (e.g., hand or facial) input to a camera, swipes to a touchscreen, or the like, to processor 714 which the programming code interprets.

The controller 706 may be a computing device such as a smart phone, a tablet, a personal diabetes controller, a dedicated diabetes therapy controller, or the like. In an example, the controller 706 may include a processor 714, a controller memory 712, a user interface 716, and a communication device 722. The controller 706 may contain analog and/or digital circuitry that may be implemented as a processor 714 for executing processes based on programming code stored in the controller memory 712, such as the medication delivery algorithm or application (MDA) 710 and/or the exercise model 782 and related programming code as well as threshold values related to exercise safety. In addition, the memory 712 may store programming code to manage a user's blood glucose levels and for controlling the delivery of the drug, medication, or therapeutic agent to the user as well as programming code that determines via the exercise model 782 whether it is safe or unsafe for a wearer (or user) to participate in exercise and cause the execution of different actions in response to the determination, as discussed above. The controller 706 may be used to program, adjust settings, and/or control operation of the wearable drug

14

delivery device 708 and/or the analyte sensor 704 as well as the optional smart accessory device 702.

The one or more transceivers, transceiver A 718 and transceiver B 720 may operate according to one or more radio-frequency protocols. In the example, the transceivers 718 and 720 may be a cellular transceiver and a Bluetooth® transceiver, respectively. For example, the transceiver A 718 or transceiver B 720 may be configured to receive and transmit signals containing information usable in the exercise model 782 by the MDA 710, such as physiological conditions of a wearer, other physiological data, IOB curves and the like.

The wearable drug delivery device 708 may include logic circuitry 748, a reservoir 738, a communication device 742, a power source 740, a memory 712, user interface (UI) 750, and a pump mechanism 744. The logic circuitry 748 may be operable to control the drug delivery device. The reservoir 738 may be configured to contain a liquid drug. The communication device 742 may be coupled to the logic circuitry 748. The pump mechanism 744 may be responsive to the logic circuitry 748 and fluidically coupled to the reservoir 738.

The memory 712 may store programming code executable by the processor 714. The programming code, for example, may enable the processor 714 to control expelling insulin from the reservoir 738 in response to control signals from the controller 706 and MDA 710 or based on signals from the optional MDA 780.

In the example, the communication device 742, which may be a receiver, a transmitter, or a transceiver that operates according to one or more radio-frequency protocols, such as Bluetooth, Wi-Fi, a near-field communication standard, a cellular standard, or the like. The communication device 764 may enable the logic circuitry 748 to communicate with the controller 706 and the analyte sensor 704.

The wearable drug delivery device 708 may be attached to the body of a user, such as a patient or diabetic, at an attachment location and may deliver any therapeutic substance to a user at or around the attachment location. For example, a bottom surface of the wearable drug delivery device 708 may include an adhesive to facilitate attachment to the skin of a wearer.

The reservoir 738 may store drugs, medications or therapeutic agents suitable for automated delivery, such as diabetes treatment drugs (e.g., insulin, glucagon, glucagon-like peptides), pain relief drugs (e.g., morphine), hormones, blood pressure medicines, chemotherapy drugs, or the like. The wearable drug delivery device 708 may include a needle or cannula (not shown in this example) coupled to the reservoir 738 and extending into the body of the wearer for delivering the drug into the wearer's body of the user (which may be done subcutaneously, intraperitoneally, or intravenously), and a pump mechanism 744 under control of the logic circuitry 748 transfers the drug from the reservoir 738 through the needle or cannula and into the wearer.

The power source 740, such as a battery, a piezoelectric device, other forms of energy harvesting devices, or the like, for supplying electrical power to the pump mechanism 744 and/or other components of the wearable drug delivery device 708.

In some examples, the wearable drug delivery device 708 may include a user interface 750, which may be a keypad, a touchscreen display, levers, light-emitting diodes, buttons on a housing of the drug delivery device 708, a microphone, a camera, a speaker, a display, or the like, that is configured to allow a user to enter information and allow the drug delivery device 708 to output information for presentation to

the user (e.g., audio, visual or vibrational alarm signals or the like). The user interface **750** may provide inputs, such as a voice input, a gesture (e.g., hand or facial) input to an optical sensor, swipes to a touchscreen, or the like, to processor **714** which the programming code interprets.

The wearable drug delivery device **708** may optionally include a device sensor **754** that may include an accelerometer, a gyroscope, a skin conductance measuring device (e.g., to measure perspiration due to exercise), or the like. Signals from the device sensor **754** may be provided to the controller **706** for use in determining whether a wearer is participating in exercise.

The smart accessory device **702** may be a smart watch, another wearable smart device, including eyeglasses, provided by other manufacturers, a global positioning system-enabled wearable device, a wearable fitness device, smart clothing, or the like, and may be operable to communicate with the other components of system **700** via wireless communication links **774**, **776**, or **778**.

For example, the smart accessory device **702** may include a communication device **736**, a processor **734**, a user interface **732** and a memory **730**. The user interface **732** may be a graphical user interface presented on a touchscreen display of the smart accessory device **702**. The memory **730** may store programming code to operate different functions of the smart accessory device **702** as well as an instance of the MDA **728**. The processor **734** that may execute programming code, such as MDA **728** for controlling the wearable drug delivery device **708** to implement the processes and techniques of FIGS. 1-6 described herein.

The analyte sensor **704** may include a processor **758**, a memory **762**, a sensing/measuring device **766**, a user interface **756**, a power source **752**, and a communication device **766**. The analyte sensor **704** may, for example, be a blood glucose monitor removably attachable via adhesive, for example, to a body of the wearer. In such an example, the analyte sensor **704** is operable to measure a blood glucose measurement value of the wearer (not shown) and communicate with the controller **706** and the drug delivery device **708** via the communication device **764** under the control of the processor **758**. The memory **730** may be configured to store information and programming code, such as an instance of the MDA **760**.

The analyte sensor **704** may be configured to detect one or more different analytes, such as lactate, ketones, uric acid, sodium, potassium, alcohol levels, blood glucose, proteins, hormones, or the like, and output results of the detections, such as measurement values or the like, for receipt by one or more of **702**, **706** or **708**. The analyte sensor **704** may, in an example, be configured to measure the blood glucose value at a predetermined time interval, such as every 5 minutes, or the like. The communication device **764** of analyte sensor **704** may have circuitry that operates as a transceiver for communicating the measured blood glucose values to the controller **706** over a wireless link **768** or with the wearable drug delivery device **708** over the wireless communication link **772**. While called an analyte sensor **704**, the sensing/measuring device **533** of the analyte sensor **704** may include one or more additional sensing elements, such as a heart rate monitor, a pressure sensor, or the like.

The processor **758** of the analyte sensor **704** may be operable to perform many functions. For example, the programming code stored in the memory **762** may enable the processor **758** to manage the collection and analysis of data detected by the sensing and measuring device **766**, such as blood glucose measurement values, providing trend information and the like. The processor **758** may include discrete,

specialized logic and/or components, an application-specific integrated circuit, a microcontroller or processor that executes software instructions, firmware, programming instructions stored in memory (such as memory **762**), or any combination thereof.

Services provided by cloud-based services **724** may include data storage that stores anonymized data, such as blood glucose measurement values, historical IOB or TDI, maximum and minimum boundary values, therapeutic exogenous substance concentration values, substance sensitivity values, and other forms of data. In addition, the cloud-based services **724** may process the anonymized data from multiple users to provide generalized information related to TDI, insulin sensitivity, IOB, and the like. The cloud-based services **724** may be accessed via data network device **786**, which may be a Wi-Fi device, a cellular communication tower, a local area network, a campus wide network or the like.

The wireless communication links **726**, **768**, **770**, **772**, **774**, and **776** may be any type of wireless link operating using known wireless communication standards or proprietary standards. As an example, the wireless communication links communication links **726**, **768**, **770**, **772**, **774**, and **776** may provide communication links based on Bluetooth®, Zigbee®, Wi-Fi, a near-field communication standard, a cellular standard, or any other wireless protocol via the respective communication devices communication links **726**, **768**, **770**, **772**, **774**, and **776**.

Software related implementations of the techniques described herein, such as the processes examples described with reference to FIGS. 1-6 may include, but are not limited to, firmware, application specific software, or any other type of computer readable instructions that may be executed by one or more processors. The computer readable instructions may be provided via non-transitory computer-readable media. Hardware related implementations of the techniques described herein may include, but are not limited to, integrated circuits (ICs), application specific ICs (ASICs), field programmable arrays (FPGAs), and/or programmable logic devices (PLDs). In some examples, the techniques described herein, and/or any system or constituent component described herein may be implemented with a processor executing computer readable instructions stored on one or more memory components.

In addition, or alternatively, while the examples may have been described with reference to a closed loop algorithmic implementation, variations of the disclosed examples may be implemented to enable open loop use. The open loop implementations allow for use of different modalities of delivery of insulin such as smart pen, syringe or the like. For example, the disclosed AP application and algorithms may be operable to perform various functions related to open loop operations, such as the generation of prompts requesting the input of information such as weight or age. Similarly, a dosage amount of insulin may be received by the AP application or algorithm from a user via a user interface. Other open-loop actions may also be implemented by adjusting user settings or the like in an AP application or algorithm.

Some examples of the disclosed device or processes may be implemented, for example, using a storage medium, a computer-readable medium, or an article of manufacture which may store an instruction or a set of instructions that, if executed by a machine (i.e., processor or controller), may cause the machine to perform a method and/or operation in accordance with examples of the disclosure. Such a machine may include, for example, any suitable processing platform,

computing platform, computing device, processing device, computing system, processing system, computer, processor, or the like, and may be implemented using any suitable combination of hardware and/or software. The computer-readable medium or article may include, for example, any suitable type of memory unit, memory, memory article, memory medium, storage device, storage article, storage medium and/or storage unit, for example, memory (including non-transitory memory), removable or non-removable media, erasable or non-erasable media, writeable or re-writable media, digital or analog media, hard disk, floppy disk, Compact Disk Read Only Memory (CD-ROM), Compact Disk Recordable (CD-R), Compact Disk Rewritable (CD-RW), optical disk, magnetic media, magneto-optical media, removable memory cards or disks, various types of Digital Versatile Disk (DVD), a tape, a cassette, or the like. The instructions may include any suitable type of code, such as source code, compiled code, interpreted code, executable code, static code, dynamic code, encrypted code, programming code, and the like, implemented using any suitable high-level, low-level, object-oriented, visual, compiled and/or interpreted programming language. The non-transitory computer readable medium embodied programming code may cause a processor when executing the programming code to perform functions, such as those described herein.

Certain examples of the present disclosure were described above. It is, however, expressly noted that the present disclosure is not limited to those examples, but rather the intention is that additions and modifications to what was expressly described herein are also included within the scope of the disclosed examples. Moreover, it is to be understood that the features of the various examples described herein were not mutually exclusive and may exist in various combinations and permutations, even if such combinations or permutations were not made express herein, without departing from the spirit and scope of the disclosed examples. In fact, variations, modifications, and other implementations of what was described herein will occur to those of ordinary skill in the art without departing from the spirit and the scope of the disclosed examples. As such, the disclosed examples are not to be defined only by the preceding illustrative description.

Program aspects of the technology may be thought of as “products” or “articles of manufacture” typically in the form of executable code and/or associated data that is carried on or embodied in a type of non-transitory, machine readable medium. Storage type media include any or all the tangible memory of the computers, processors or the like, or associated modules thereof, such as various semiconductor memories, tape drives, disk drives and the like, which may provide non-transitory storage at any time for the software programming. It is emphasized that the Abstract of the Disclosure is provided to allow a reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, various features are grouped together in a single example for streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate example. In the appended claims, the terms “including” and “in which” are used as the plain-English equiva-

lents of the respective terms “comprising” and “wherein,” respectively. Moreover, the terms “first,” “second,” “third,” and so forth, are used merely as labels and are not intended to impose numerical requirements on their objects.

The foregoing description of examples has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed. Many modifications and variations are possible considering this disclosure. It is intended that the scope of the present disclosure be limited not by this detailed description, but rather by the claims appended hereto. Future filed applications claiming priority to this application may claim the disclosed subject matter in a different manner and may generally include any set of one or more limitations as variously disclosed or otherwise demonstrated herein.

What is claimed is:

1. A wearable drug delivery device, comprising:

a processor,
a reservoir configured to contain insulin;
a pump mechanism communicatively coupled to the processor and fluidically coupled to the reservoir; and
a memory coupled to the processor and operable to store programming code that, when executed by the processor, causes the processor to:
use an insulin on board (IOB) curve in determining a basal dose of insulin to deliver to a wearer;
receive data indicative of a physiological condition of the wearer of the wearable drug delivery device;
evaluate the received data with relation to the wearer participating in exercise;
based on a result of the evaluation,
determine whether it is safe for the wearer to participate in exercise; and
output an exercise safety signal indicating a result of the determination of whether it is safe or unsafe for the wearer to participate in exercise; and
where the evaluation determines that the wearer is exercising,
lengthening the IOB curve, and
using the lengthened IOB curve to determine a next basal dose of insulin for the wearer.

2. The wearable drug delivery device of claim 1, wherein the processor, when determining whether it is safe for the wearer to participate in exercise, is further operable to:
determine whether the result of the evaluation indicates a presence of one or more conditions; and
based on the determination of the presence of the one or more conditions, generate an indication that exercise is unsafe.

3. The wearable drug delivery device of claim 2, wherein the one or more conditions include a blood glucose measurement value, a total daily insulin amount, a request for a bolus or a request to use a hypoglycemia protect mode, a current calculation of an amount of insulin onboard, or an amount of insulin delivered in a past evaluation period.

4. The wearable drug delivery device of claim 1, wherein the processor, when determining whether it is safe for the wearer to participate in exercise, is further operable to:
determine whether the result of the evaluation indicates that a trajectory of blood glucose measurement values of the wearer is trending downward; and
based on the determination that the trajectory of the glucose measurement values of the wearer are trending downward, generate an indication that exercise is unsafe.

19

5. The wearable drug delivery device of claim 1, wherein the processor, when executing the stored programming code, is further operable to:

retrieve physiological data from the memory, wherein the physiological data contains a number of days of data related to the wearer's participation in exercise; and update an exercise model using the retrieved physiological data.

6. The wearable drug delivery device of claim 5, wherein the processor, when evaluating the received data related to the wearer participating in exercise, is further operable to: input values obtained from the received data in the exercise model; and

analyze an output of the exercise model with reference to threshold values related to exercise safety for the wearer.

7. The wearable drug delivery device of claim 1, wherein the received data includes a current blood glucose measurement value, a total daily insulin value, a time and amount of a requested bolus, a current insulin onboard value, or an amount of insulin delivered within a past evaluation period.

8. The wearable drug delivery device of claim 7, wherein the past evaluation period is a 15 minute period, a 30 minute period, a 60 minute period or a 120 minute period of time.

9. The wearable drug delivery device of claim 1, further comprising:

a communication device, wherein the communication device is operable to provide a wireless communication connection with an external device; and

the processor is further operable to:

control the communication device;

establish the wireless communication connection with the external device,

wherein the external device is a data network device;

retrieve physiological data from a memory source, wherein the physiological data contains a number of days of data related to the wearer's participation in exercise; and

update an exercise model using the retrieved physiological data.

10. The wearable drug delivery device of claim 1, wherein the processor is further configured to:

obtain data from a dataset including physiological data related to physiological conditions of the wearer;

split data in the dataset into training data and validation data;

train a logistic regression model using the training data; use the validation data to validate an output of the logistic regression model;

based on a result of the validation of the output of the logistic regression model, perform hyperparameter tuning to improve accuracy of the logistic regression model;

retrieve the obtained data related to the physiological conditions of the wearer;

input the retrieved data into the logistic regression model for evaluation;

based on a value of output data from the logistic regression model, label the output data as either indicating an unsafe condition for exercise or a safe condition for exercise; and

generate the exercise safety signal based on any label indicating the unsafe condition for exercise.

11. The wearable drug delivery device of claim 1, further comprising:

an output device coupled to the processor, wherein the processor is further operable to:

20

interpret the outputted exercise safety signal; and provide an output signal to the output device based on the outputted exercise safety signal.

12. The wearable drug delivery device of claim 1, wherein the processor is further operable to:

modify a medication treatment plan based on the outputted exercise safety signal.

13. A wearable drug delivery system, comprising:

a drug delivery device, including:

logic circuitry operable to control the drug delivery device;

a reservoir configured to contain insulin;

a communication device coupled to the logic circuitry; and

a pump mechanism responsive to the logic circuitry and fluidically coupled to the reservoir; and

a controller, including:

a processor;

a memory coupled to the processor and configured to store a medication delivery application, an exercise safety application, and programming code that, when the exercise safety application is executed by the processor, the processor is operable to:

receive an input related to a physiological condition of a wearer, wherein the input is from the wearer;

evaluate data parameters related to physiological conditions of a wearer and the received input related to the physiological condition of a wearer;

determine whether the wearer has participated in exercise;

in response to a determination that the wearer has participated in exercise and a result of the evaluation of the data parameters, generate an exercise safety signal;

present an indication of a result of an interpretation of the exercise safety signal, wherein the indication is a safe or unsafe indicator;

modify a medication treatment plan in response to the generation of the exercise safety signal, wherein the modifying includes lengthening an insulin on board (IOB) curve and using the lengthened IOB curve to determine basal insulin doses for a period of multiple hours.

14. The wearable drug delivery system of claim 13, wherein the data parameters related to physiological conditions of a wearer include one or more of:

a blood glucose measurement value,

a total daily insulin amount,

a request for a bolus,

a request to use a hypoglycemia protect mode,

a current calculation of an amount of insulin onboard, or an amount of insulin delivered in a past evaluation period.

15. The wearable drug delivery system of claim 13, wherein the controller is a device separate from the drug delivery device.

16. The wearable drug delivery system of claim 13, wherein the controller further comprises:

a user interface communicatively coupled to the processor and operable to receive the input related to a physiological condition of a wearer and provide the input to the processor, wherein the input is a request for a bolus dosage.

17. The wearable drug delivery system of claim 13, further comprising:

21

an analyte sensor configured to be attached to a body of the wearer, and including an analyte processor and an analyte communication device, wherein the analyte sensor is operable to:

measure a blood glucose measurement value of the wearer, and

communicate with the controller and the drug delivery device via the analyte communication device under control of the analyte processor.

18. The wearable drug delivery system of claim 13, wherein the processor, when evaluating the data parameters, is further operable to:

input data parameters including a blood glucose measurement value, a total daily insulin value, a time and an amount of a requested bolus, an amount of insulin onboard value, or an amount of insulin delivered within a past evaluation period into an exercise model; and based on a result of an output from the exercise model, generate the exercise safety signal.

19. A non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by a processor, cause the processor to:

22

determine that wearer of a wearable drug delivery device is participating in exercise;

lengthen an insulin on board curve in response to the determination that a wearer is participating in exercise; use the lengthened insulin on board curve to calculate subsequent doses of insulin to be delivered to the wearer during a period of multiple hours.

20. The non-transitory computer-readable storage medium of claim 19, further including instructions that when executed by the processor, cause the processor to:

in response to the determination that the wearer of a wearable drug delivery device is participating in exercise, determine a current insulin onboard of the wearer; compare the determined current insulin onboard to a threshold value set for the wearer;

in response to the current insulin onboard being above the threshold value set for the wearer, generate an exercise safety signal indicating an unsafe exercise condition; and

based on the generated exercise safety signal, suspend delivery of insulin for a period of time.

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